

Dining Philosophers Problem

The Dining Philosophers problem demonstrates **process synchronization** when multiple philosophers share limited resources (forks). Five philosophers repeatedly **think, become hungry, acquire two forks, eat, and release the forks**. In our implementation, **semaphores** control access to the forks, and **resource ordering** is used by always acquiring the lower-numbered fork first. This prevents **deadlock** by eliminating circular wait. The program successfully allows all five philosophers to complete their eating cycles without getting permanently blocked.

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shaurya@Shaurya6480:~/xv6-peterson$ make qemu CPUS=2
xv6 kernel is booting

hart 1 starting
init: starting sh
$ dining
Dining Philosophers started
Using resource ordering for deadlock avoidance
Each philosopher will eat 5 times

Philosopher 0: THINKING
Philosopher 1: THINKING
Philosopher 0: THINKING -> HUNGRY
Philosopher 1: THINKING -> HUNGRY
Philosopher 0: HUNGRY -> EATING
Philosopher 2: THINKING
Philosopher 1: HUNGRY -> EATING
Philosopher 0: EATING -> THINKING
Philosopher 1: EATING -> THINKING
Philosopher 0: THINKING
Philosopher 1: THINKING
Philosopher 0: THINKING -> HUNGRY
Philosopher 1: THINKING -> HUNGRY
Philosopher 0: HUNGRY -> EATING
Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING
Philosopher 0: EATING -> THINKING
Philosopher 1: HUNGRY -> EATING
Philosopher 0: THINKING
Philosopher 2: HUNGRY -> EATING
Philosopher 0: THINKING -> HUNGRY
Philosopher 1: EATING -> THINKING
Philosopher 0: HUNGRY -> EATING
Philosopher 1: THINKING
Philosopher 0: EATING -> THINKING
Philosopher 1: THINKING -> HUNGRY
Philosopher 0: THINKING
Philosopher 0: THINKING -> HUNGRY
Philosopher 1: HUNGRY -> EATING
Philosopher 2: EATING -> THINKING
Philosopher 1: EATING -> THINKING
Philosopher 0: HUNGRY -> EATING
Philosopher 1: THINKING
Philosopher 2: THINKING
Philosopher 0: EATING -> THINKING
Philosopher 1: THINKING -> HUNGRY
Philosopher 0: THINKING
Philosopher 1: HUNGRY -> EATING
Philosopher 0: THINKING -> HUNGRY
Philosopher 1: EATING -> THINKING
Philosopher 0: HUNGRY -> EATING
Philosopher 1: THINKING
Philosopher 0: EATING -> THINKING
Philosopher 1: THINKING -> HUNGRY
Philosopher 0 completed all 5 cycles
Philosopher 1: HUNGRY -> EATING
Philosopher 2: THINKING -> HUNGRY
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Philosopher 1: THINKING
Philosopher 2: THINKING
Philosopher 0: EATING -> THINKING
Philosopher 1: THINKING -> HUNGRY
Philosopher 0: THINKING
Philosopher 1: HUNGRY -> EATING
Philosopher 0: THINKING -> HUNGRY
Philosopher 1: EATING -> THINKING
Philosopher 0: HUNGRY -> EATING
Philosopher 1: THINKING
Philosopher 0: EATING -> THINKING
Philosopher 1: THINKING -> HUNGRY
Philosopher 0 completed all 5 cycles
Philosopher 1: HUNGRY -> EATING
Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING -> HUNGRY
Philosopher 1: EATING -> THINKING
Philosopher 2: HUNGRY -> EATING
Philosopher 1 completed all 5 cycles
Philosopher 2: EATING -> THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: THINKING
Philosopher 3: EATING -> THINKING
Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING
Philosopher 2: HUNGRY -> EATING
Philosopher 3: THINKING -> HUNGRY
Philosopher 2: EATING -> THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: THINKING
Philosopher 3: EATING -> THINKING
Philosopher 3: THINKING
Philosopher 2: HUNGRY -> EATING
Philosopher 3: THINKING -> HUNGRY
Philosopher 2: EATING -> THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: THINKING
Philosopher 3: EATING -> THINKING
Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING
Philosopher 3: THINKING -> HUNGRY
Philosopher 2: HUNGRY -> EATING
Philosopher 4: THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: EATING -> THINKING
Philosopher 3: EATING -> THINKING
Philosopher 2 completed all 5 cycles
Philosopher 3: THINKING
Philosopher 4: THINKING -> HUNGRY
Philosopher 3: THINKING -> HUNGRY
Philosopher 4: HUNGRY -> EATING
Philosopher 3: HUNGRY -> EATING
Philosopher 4: EATING -> THINKING
Philosopher 3: EATING -> THINKING
Philosopher 4: THINKING

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Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING
Philosopher 2: HUNGRY -> EATING
Philosopher 3: THINKING -> HUNGRY
Philosopher 2: EATING -> THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: THINKING
Philosopher 3: EATING -> THINKING
Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING
Philosopher 2: HUNGRY -> EATING
Philosopher 3: THINKING -> HUNGRY
Philosopher 2: EATING -> THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: THINKING
Philosopher 3: EATING -> THINKING
Philosopher 2: THINKING -> HUNGRY
Philosopher 3: THINKING
Philosopher 3: THINKING -> HUNGRY
Philosopher 2: HUNGRY -> EATING
Philosopher 4: THINKING
Philosopher 3: HUNGRY -> EATING
Philosopher 2: EATING -> THINKING
Philosopher 3: EATING -> THINKING
Philosopher 2 completed all 5 cycles
Philosopher 3: THINKING
Philosopher 4: THINKING -> HUNGRY
Philosopher 3: THINKING -> HUNGRY
Philosopher 4: HUNGRY -> EATING
Philosopher 3: HUNGRY -> EATING
Philosopher 4: EATING -> THINKING
Philosopher 3: EATING -> THINKING
Philosopher 4: THINKING
Philosopher 3 completed all 5 cycles
Philosopher 4: THINKING -> HUNGRY
Philosopher 4: HUNGRY -> EATING
Philosopher 4: EATING -> THINKING
Philosopher 4: THINKING
Philosopher 4: THINKING -> HUNGRY
Philosopher 4: THINKING -> HUNGRY
Philosopher 4: HUNGRY -> EATING
Philosopher 4: EATING -> THINKING
Philosopher 4: THINKING
Philosopher 4: THINKING -> HUNGRY
Philosopher 4: HUNGRY -> EATING
Philosopher 4: EATING -> THINKING
Philosopher 4 completed all 5 cycles

All philosophers completed successfully
Dining Philosophers completed

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